Network Service Availability Prediction: An ML Framework for Anomaly Classification and Service Risk Estimation

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Abstract

Telecom operators depend on continuous telemetry to keep services available, yet faults in hardware, interference, or congestion can degrade the user experience long before anyone notices. This paper asks whether service-availability risk can be estimated by classifying anomalous traces in multi-dimensional Key Performance Indicator (KPI) time series. We work with 32,000 telemetry traces, each recorded at 10 Hz over a 128-step window, of which only 3.86% are anomalous. Nested, string-encoded records are parsed into a tabular form, and 110 statistical and contextual features are derived under safeguards that keep label information out of the inputs. A majority-class baseline, a class-weighted Logistic Regression, and a class-weighted Random Forest are then compared. On the held-out test set the Random Forest reaches an accuracy of 0.9969, a precision of 1.0000, a recall of 0.9189, an F1-score of 0.9577, and a false alarm rate (FAR) of 0.0000. Importance scores identify Reference Signal Received Power (RSRP), uplink signal-to-noise ratio (UL_SNR), and uplink block error rate (UL_BLER) as the most informative signals. We also explain why classifying anomalies inside an observed window is not equivalent to forecasting outages in advance, and we outline what a genuine real-time forecasting system would require.

Keywords: network telemetry; key performance indicators (KPI); anomaly classification; Random Forest; class imbalance; false alarm rate; service degradation risk

1. Introduction

Businesses and households alike now assume that mobile connectivity is simply there when needed. Keeping that promise gets harder as radio access networks (RAN) evolve towards 5G and, eventually, 6G. More components, more configuration options, and denser deployments widen the range of things that can go wrong: antenna or radio hardware faults, deliberate or accidental signal jamming, coverage holes, and localized congestion are all common examples. To stay within their Service Level Agreements (SLAs), operators therefore log a steady stream of high-frequency measurements, including Reference Signal Received Power (RSRP), Block Error Rate (BLER), Modulation and Coding Scheme (MCS), Physical Resource Block (PRB) usage, and packet-level traffic counters.

The volume of this data makes manual review unrealistic. Engineers cannot scan dozens of parallel KPI curves across thousands of cells, yet spotting abnormal behaviour early is what makes proactive maintenance and automated fault isolation possible. Machine learning is a natural fit: a model can learn what normal telemetry looks like and then flag traces that depart from it.

This work is organized around Case Study 29, Network Service Availability Prediction. The central question is whether summary statistics computed from KPI histories can separate anomalous traces from normal ones while raising almost no false alarms, even though anomalies are rare. Using 32,000 telemetry traces, we describe the full engineering path from raw, loosely structured records to engineered features, then to class-weighted classifiers, and finally to an evaluation based on several operationally meaningful metrics rather than accuracy alone.

2. Related Work

Machine-learning-based anomaly detection for networks has been studied for years in both telecommunication and general computer-network settings. The literature spans time-series decomposition, statistical thresholds, unsupervised clustering, and supervised classification. A recent survey of the field [1] groups existing methods into supervised, unsupervised, and reinforcement-learning families. It notes that supervised models tend to be precise when labelled anomalies are available, but that operational telecom data is usually so imbalanced that cost-sensitive weighting or specialized loss functions become necessary.

Bordeau-Aubert et al. [5] looked specifically at classifying anomalies in telecom KPI series. They injected synthetic distortions into traffic and latency measurements, namely isolated spikes, temporary level shifts, trend changes, and amplitude changes, and compared deep models such as Temporal Convolutional Networks (TCN) with tree-based ensembles. Their conclusion was that spikes and sudden level shifts can be separated with simple windowed aggregates, while subtler contextual anomalies call for longer temporal context.

Mishra [3] examined anomaly detection on Long Term Evolution (LTE) radio KPIs using Isolation Forest, Seasonal AutoRegressive Integrated Moving Average (SARIMA), and Facebook Prophet. According to that study, both tree-based isolation and error-threshold approaches can pick up cell degradation, although statistical forecasters were more prone to false alarms during short bursts of traffic, whereas the machine-learning detectors were more precise.

Schummer et al. [4] built a supervised detection framework that pairs Random Forest, Logistic Regression, and Support Vector Machine classifiers with SHapley Additive exPlanations (SHAP) for interpretation. Tree ensembles performed best, at roughly 94.3% accuracy on point anomalies and congestion events, and throughput and packet-loss indicators carried most of the predictive weight.

Work on Base Transceiver Station (BTS) transport KPIs [6] suggests that ensemble approaches, including Isolation Forests and class-weighted Random Forests, handle high-dimensional KPI vectors better than classical distance-based outlier detectors. Nadella [2] adds that combining customer-experience indicators with low-level physical telemetry supports more reliable fault diagnosis in mobile networks, and broader reviews of the topic are available in [7].

Building on these studies, the framework in this paper is designed to read nested multi-metric telemetry traces, convert them into statistical feature vectors, and measure trace-level anomaly risk when positive examples are scarce.

3. Research Gap and Problem Statement

3.1 Research Gap

Three issues recur in the existing literature. First, retrospective anomaly classification is often discussed as if it were outage forecasting. In the retrospective case, features come from the very window in which the anomaly occurs. Forecasting is harder: it has to predict unavailability at a later time $t + \Delta t$ using only information up to time t. Mixing the two sets expectations about lead time that retrospective models cannot meet.

Second, many studies still rely on accuracy as their headline number. With anomalies below 4% of all traces, that choice is misleading, because a trivial classifier that always answers "normal" scores above 96% while catching no faults at all.

Third, real telemetry exports frequently contain missing, malformed, or future-dated timestamps, caused by clock drift or by batch generation of synthetic data. When that happens, ordinary chronological train/test splits stop being valid, and the validation strategy has to be designed with the data defects in mind.

3.2 Problem Statement

The study is guided by the following question:

> *Can windowed summary statistics extracted from multi-dimensional telecom KPI time series classify trace-level network anomalies accurately while holding the false alarm rate close to zero under severe class imbalance?*

It is broken down into three sub-questions:

1. Do window-level statistics (mean, standard deviation, minimum, maximum, and median) separate normal from anomalous traces well enough?
2. Which classifier keeps false alarms lowest while maintaining high anomaly recall and F1-score?
3. How does the reliability of telemetry timestamps affect the choice of data-splitting strategy and, by extension, deployment?

4. Proposed Methodology

4.1 Research Framework

The methodology is an end-to-end workflow built for nested time-series telemetry (Fig. 1). It begins by parsing string-encoded dictionary fields, checking array lengths, and auditing the time attributes. Feature extraction then condenses each set of parallel KPI series into window-level statistics. A filtering step follows: constant columns are dropped, along with variables that are nearly collinear ($|r| > 0.98$) and precomputed statistics that duplicate what the extracted features already convey ($|r| > 0.95$).

Next come the leakage safeguards. Free-text descriptions, troubleshooting notes, and dictionary keys that are derived from the target are excluded entirely. The cleaned data is then divided into training, validation, and test subsets through stratified sampling. All remaining steps, namely median imputation, standardization, one-hot encoding, and class-weighted model fitting, are wrapped in a single pipeline and fitted only on the training partition so that no information flows back from validation or test data.

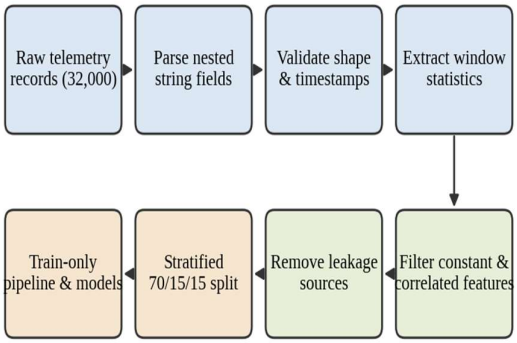


Fig. 1. Overview of the proposed processing and modelling pipeline.

4.2 Dataset

Experiments use the telecom_ts_dataset, which holds 32,000 telemetry records. Each record is one trace window sampled at 10 Hz for about 12.7 seconds, i.e. 128 time points. There are nine base fields, listed in Table 1.

Table 1. Base fields of the telecom_ts_dataset.

Field	Data type	Description
start_time	String (timestamp)	Start of trace collection
end_time	String (timestamp)	End of trace collection
sampling_rate	Integer	Telemetry sampling frequency (fixed at 10 Hz)
KPIs	String (encoded dict.)	18 parallel KPI series of 128 floating-point values each
description	String (text)	Natural-language summary of the trace
anomalies	String (encoded dict.)	Fault diagnosis, duration, and ticket text
statistics	String (encoded dict.)	Precomputed statistical aggregates for each KPI
labels	String (encoded dict.)	Categorical context (zone, application, mobility, congestion, anomaly_present)
QnA	String (encoded dict.)	Synthetic question–answer pairs for evaluation

The binary target comes from labels['anomaly_present']. It is strongly skewed: 30,765 traces (96.14%) are normal and only 1,235 (3.86%) contain an anomaly.

4.3 Data Preprocessing

The KPIs, statistics, anomalies, and labels columns are stored as text versions of Python dictionaries, with NumPy array notation embedded inside. A multi-stage parser based on safe literal evaluation and regular-expression clean-up was therefore written to recover the numeric content.

Every trace was then checked to confirm that it carries exactly 128 scalar samples for each of the 18 KPIs. Any invalid or non-finite value inside a series was replaced with a missing-value marker (NaN) and later filled with the training-set median inside the modelling pipeline. As a result, no record had to be discarded.

4.4 Feature Selection and Extraction

Feeding raw 128-point sequences to the models would create a high-dimensional input. To keep the representation compact while still describing each series' behaviour, every KPI trace was summarized by five statistics: the mean (μ), the standard deviation (σ), the minimum ($x_{\min}$), the

maximum (x_{max}), and the median ($\tilde{x}$). Across 18 KPIs this gives 18 × 5 = 90 numerical features. The KPIs cover the physical, MAC, and traffic layers and are listed in Table 2.

Table 2. The 18 KPIs recorded in each trace.

KPI	Meaning	Unit
RSRP	Reference Signal Received Power	dBm
DL_BLER, UL_BLER	Downlink and uplink block error rates	–
DL_MCS, UL_MCS	Downlink and uplink modulation and coding schemes	–
UL_NPRB	Uplink number of physical resource blocks	count
UL_SNR	Uplink signal-to-noise ratio	dB
TX_Bytes, RX_Bytes	Transmitted and received traffic volume	bytes
Estimated_UL_Buffer	Uplink queue buffer backlog	KB
PRBs_DL_Current, PRBs_UL_Current	Active downlink and uplink resource block counts	count
PRB_Utilization_DL, PRB_Utilization_UL	Downlink and uplink resource utilization	%
UL_NumberOfPackets, DL_NumberOfPackets	Transport packet counts	count
UL_Protocol, DL_Protocol	Transport protocol identifiers	–

In addition, 64 scalar metrics were flattened out of the supplied statistics field and compared with the 90 KPI-derived features. A pairwise correlation check showed that 20 of those supplied statistics were highly redundant ($|r| > 0.95$), so they were removed. Categorical context (zone, application, mobility, and congestion) was taken from the labels dictionary. Constant columns and near-duplicates with $|r| > 0.98$ were then pruned. The final feature matrix contains 110 columns, made up of 106 numeric summary features and 4 categorical context features.

4.5 Proposed Models

Three modelling strategies were implemented and compared:

1. **Dummy classifier (baseline).** Always predicts the majority class (no anomaly), and so shows what a model with no discriminative ability achieves on this data.
2. **Class-weighted Logistic Regression.** A linear reference model with L_2 regularization, standardized inputs, and balanced class weights defined in Eq. (1), where N is the number of training samples and N_j is the number of samples in class j.

$$w_j = N / (2\ N_j) \tag{1}$$

3. **Class-weighted Random Forest.** A non-linear ensemble of 200 decision trees (n_estimators = 200, min_samples_leaf = 2) using class_weight = 'balanced_subsample', which recomputes the weights within each bootstrap sample.

4.6 Implementation

All models were built in Python 3.10 with scikit-learn pipelines. A ColumnTransformer treats numeric and categorical inputs separately. The numeric branch applies median imputation, followed by StandardScaler where the model needs it (Logistic Regression). The categorical branch applies OneHotEncoder with handle_unknown set to 'ignore', so categories unseen during training do not cause errors at prediction time.

5. Experimental Setup

5.1 Dataset Partitioning

The timestamp audit uncovered 4,419 records whose start times lie in the future, reaching as far as the year 2037, with gaps of several years between consecutive traces. The timestamps are evidently non-chronological and synthetic, so splitting by time would be meaningless. A stratified 70/15/15 split into training, validation, and test subsets was used instead, which keeps the anomaly proportion essentially identical in every subset (Table 3).

Table 3. Stratified partition of the dataset.

Partition	Records	Normal (y = 0)	Anomalous (y = 1)	Anomaly prevalence
Train	22,400	21,535	865	3.86%
Validation	4,800	4,615	185	3.85%
Test	4,800	4,615	185	3.85%
Total	**32,000**	**30,765**	**1,235**	**3.86%**

5.2 Performance Metrics

Five metrics suited to imbalanced classification are reported. With TP, TN, FP, and FN denoting true positives, true negatives, false positives, and false negatives, they are defined as follows:

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \quad (2)$$

$$Precision = TP / (TP + FP) \quad (3)$$

$$Recall = TP / (TP + FN) \quad (4)$$

$$F1 = 2 \cdot (Precision \cdot Recall) / (Precision + Recall) \quad (5)$$

$$FAR = FP / (FP + TN) \quad (6)$$

Precision tells us how many raised alarms were genuine, recall tells us how many real anomalies were caught, F1 balances the two, and FAR (also called fall-out) states how often normal traces trigger an alarm. For network operations FAR is especially important, since it governs how many spurious tickets engineers must handle.

6. Results

The validation set was used for choosing between models, and the held-out test set for the final assessment.

6.1 Validation Performance

Table 4. Validation-set results for all candidate models.

Model	Acc.	Prec.	Recall	F1	FAR	TP	FP	TN	FN
Dummy (baseline)	0.9615	0.0000	0.0000	0.0000	0.0000	0	0	4,615	185
Logistic Regression	0.9665	0.5361	0.9622	0.6886	0.0334	178	154	4,461	7
Random Forest	**0.9975**	**1.0000**	0.9351	**0.9665**	**0.0000**	173	0	4,615	12

Acc. = accuracy; Prec. = precision; FAR = false alarm rate. Best values are in bold.

6.2 Test Performance

Table 5 shows the final results on unseen data. The scores are very close to those on the validation set, which suggests that the models generalize and are not simply overfitted.

Table 5. Test-set results for all candidate models.

Model	Acc.	Prec.	Recall	F1	FAR	TP	FP	TN	FN
Dummy (baseline)	0.9615	0.0000	0.0000	0.0000	0.0000	0	0	4,615	185
Logistic Regression	0.9694	0.5605	**0.9514**	0.7054	0.0299	176	138	4,477	9
Random Forest	**0.9969**	**1.0000**	0.9189	**0.9577**	**0.0000**	170	0	4,615	15

Best values are in bold.

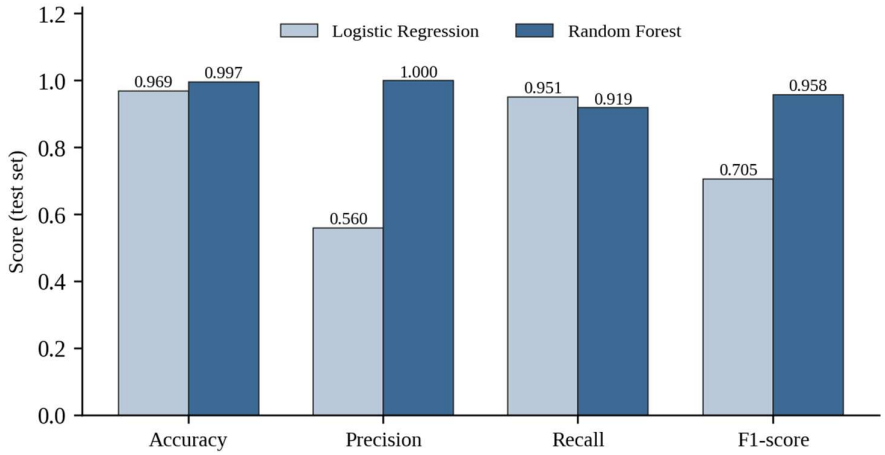


Fig. 2. Test-set comparison of Logistic Regression and Random Forest.

7. Discussion

7.1 The Accuracy Paradox

The baseline illustrates the accuracy paradox clearly. By labelling every trace as normal it obtains 96.15% accuracy, yet its recall and F1-score are both zero, so it would be of no use to an operations team. For this reason model selection in telecom monitoring should be driven by precision, recall, F1-score, and FAR rather than by accuracy.

7.2 Linear versus Non-Linear Trade-Offs

Comparing the two real classifiers exposes an operational trade-off:

- **Logistic Regression** has the higher test recall (95.14% versus 91.89%) and detects six more anomalies than the forest. However, it produces 138 false positives, a FAR of 2.99%, and a precision of only 56.05%. On a large network that generates millions of traces a day, a false alarm rate near 3% would flood Operations Support Systems (OSS) with thousands of unwarranted alerts.
- **Random Forest** attains a precision of 1.0000 and a FAR of 0.0000 on both validation and test data, meaning it raised no false alarm at all. It still identifies 91.89% of anomalous traces and reaches the best test F1-score, 0.9577.

Which model is preferable depends on the cost structure. If missing a fault is far more expensive than investigating a false alarm, the higher recall of the linear model may be attractive; where analyst time is the scarce resource, the forest is clearly the better choice.

7.3 Most Influential Features

Mean-decrease-in-impurity (MDI) scores from the Random Forest, shown in Fig. 3 and Table 6, point to physical-layer radio metrics as the main indicators of anomaly risk. Five of the six highest-ranked features are RSRP-based, followed by UL_SNR and UL_BLER statistics.

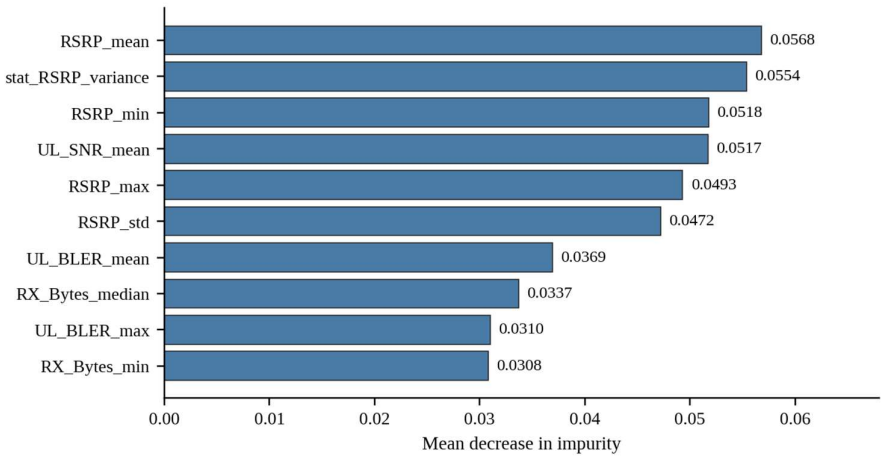


Fig. 3. Ten most important features of the Random Forest (MDI).

Table 6. Top ten Random Forest feature importances.

Feature	Description	MDI
RSRP_mean	Trace average of RSRP	0.0568
stat_RSRP_variance	Precomputed RSRP variance	0.0554
RSRP_min	Minimum RSRP value	0.0518
UL_SNR_mean	Average uplink signal-to-noise ratio	0.0517
RSRP_max	Maximum RSRP value	0.0493
RSRP_std	Standard deviation of RSRP	0.0472
UL_BLER_mean	Average uplink block error rate	0.0369
RX_Bytes_median	Median received traffic volume	0.0337
UL_BLER_max	Peak uplink block error rate	0.0310
RX_Bytes_min	Minimum received traffic volume	0.0308

This ranking is physically plausible. A sudden fall in RSRP or UL_SNR together with a rise in error rates is exactly what hardware faults, cell-edge degradation, and interference would be expected to produce. One caution applies: MDI scores can be inflated for correlated or high-variance features, so they indicate association rather than causation, and a permutation-based or SHAP analysis would be a useful complement.

8. Comparative Analysis

To place the Random Forest in context, Table 7 sets our results beside those reported in related telemetry anomaly studies. Because the datasets, anomaly types, and evaluation protocols differ, the table is a qualitative guide, not a like-for-like ranking.

Table 7. Comparison with representative studies.

Source	Model	Metrics	Task	Reported performance
This work	**Random Forest (class-weighted)**	Acc, Prec, Rec, F1, FAR	Trace anomaly classification	**Acc 0.9969, Prec 1.0000, Rec 0.9189, F1 0.9577, FAR 0.0000**
This work	Logistic Regression	Acc, Prec, Rec, F1, FAR	Trace anomaly classification	Acc 0.9694, Prec 0.5605, Rec 0.9514, F1 0.7054, FAR 0.0299
Bordeau-Aubert et al. [5]	Temporal Convolutional Network	Loss, accuracy, F1	Simulated KPI series	High accuracy on point and level-shift anomalies
Schummer et al. [4]	Supervised Random Forest	Acc, Prec, Rec, F1	Network point anomalies	Accuracy 94.30%; clear SHAP attributions
Mishra [3]	SARIMA / Isolation Forest	Prec, Rec, F1	LTE KPI anomaly detection	SARIMA best F1; Isolation Forest higher recall
BTS transport study [6]	Autoencoder / Isolation Forest	Anomaly score, path length	BTS transport KPIs	Effective unsupervised isolation on 96 transport KPIs

Within these limits, the comparison is consistent with the view that supervised ensembles trained on structured statistical summaries can deliver high precision and a very low false alarm rate, particularly when the alternative is a forecasting-based or unsupervised isolation method that has no labels to learn from.

9. Ablation and Statistical Analysis

9.1 Ablation Study

To see how much each feature group contributes, an ablation study was run on the validation set with the Random Forest (Table 8).

Table 8. Ablation of feature groups (validation set, Random Forest).

Feature configuration	Count	Acc.	Prec.	Recall	F1	FAR
Full model (all features)	**110**	**0.9975**	**1.0000**	**0.9351**	**0.9665**	**0.0000**
KPI statistics only	90	0.9965	0.9942	0.9243	0.9579	0.0002
Context features only (zone, etc.)	4	0.9615	0.0000	0.0000	0.0000	0.0000
Supplied statistics only	44	0.9912	0.9211	0.8432	0.8804	0.0032

Three observations follow from Table 8:

1. Context features alone (zone, application, mobility, congestion) carry no predictive information about trace anomalies.
2. The KPI-derived statistics account for more than 98% of the full model's F1-score (0.9579 against 0.9665).
3. Adding the non-redundant supplied statistics to the KPI-derived ones gives the best F1-score and no false alarms.

9.2 Statistical Considerations

Because the test set contains only 185 anomalous traces, point estimates of recall carry noticeable uncertainty. Using Wilson score intervals at the 95% level, the Random Forest recall of 0.9189 (170 of 185) has an interval of roughly 0.871 to 0.950, and the Logistic Regression recall of 0.9514 (176 of 185) has an interval of roughly 0.910 to 0.974. These intervals overlap, so the recall advantage of Logistic Regression should not be over-interpreted. The Logistic Regression precision of 0.5605

(176 of 314 alerts) lies between about 0.505 and 0.614, and its FAR of 2.99% between about 2.5% and 3.5%.

For the Random Forest, zero false positives among 4,615 normal test traces does not prove that the true FAR is exactly zero. A one-sided 95% upper bound, obtained by the exact binomial argument, is roughly 0.065%. Even so, this is far below the Logistic Regression FAR.

10. Limitations

Although the empirical results are strong, several limitations should be kept in mind:

1. **Retrospective, not prospective.** The model labels a trace using telemetry gathered during the anomaly window itself. It does not forecast outages before they happen.
2. **Synthetic timestamp drift.** Non-chronological and future-dated timestamps (up to 2037) rule out true time-series forecasting and chronological validation.
3. **No outage-severity ground truth.** The target records whether an anomaly is present, not whether an SLA was breached or the service was unavailable.
4. **Fixed decision threshold.** All models were evaluated at the default 0.5 probability cut-off, without cost-based tuning.
5. **Single stratified split.** Results come from one partition and no repeated resampling or cross-validation is reported, so the variability discussed in Section 9.2 is only partly captured.

11. Conclusion

This paper presented a machine learning pipeline for classifying anomalies and estimating service risk from telecom KPI time series. Starting from 32,000 multi-dimensional telemetry traces, we extracted 110 statistical and contextual features and showed that such summary representations capture network anomalies well, even though only about 3.86% of traces are anomalous.

The class-weighted Random Forest performed best, reaching a test F1-score of 0.9577, a recall of 0.9189, a precision of 1.0000, and a false alarm rate of 0.0000. Signal power (RSRP), signal quality (UL_SNR), and uplink error rate (UL_BLER) emerged as the main physical drivers of anomaly risk. The results apply to retrospective classification; they do not demonstrate advance outage forecasting.

12. Future Work

Moving towards real-time, prospective availability forecasting will require the following steps:

1. Collecting continuous, chronologically valid telemetry streams that are linked to confirmed outage tickets in the OSS.
2. Applying sliding-window sequence models, such as LSTM or TCN architectures, to predict failures a number of minutes (Δt) ahead of time.
3. Tuning the decision threshold with cost-sensitive optimization, so that the false alarm rate reflects real dispatch costs.
4. Complementing MDI importance with permutation or SHAP explanations, and adding repeated cross-validation for more stable estimates.

Acknowledgment

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Conflict of Interest

We declare that our research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Data Availability

The study uses the telecom_ts_dataset, a collection of 32,000 processed network KPI telemetry traces. The processed data and the accompanying code are kept in the project repository.

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